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A palaeomagnetic reconnaissance of Kashmir, northwestern Himalaya, India

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Thermal demagnetization studies of 800 Palaeozoic and Lower Mesozoic samples, mainly carbonates, from 14 localities in Kashmir, showed four magnetic components:

(A) A recent field component of normal polarity and large intensity.

(B) A Middle to Late Tertiary secondary component of large intensity and of reversed and normal polarity, which indicates clockwise rotation of central and eastern Kashmir over approximately 45° with respect to peninsular Indo-Pakistan.

(C) Primary magnetic components of Triassic, Late Permian, Early Carboniferous, Siluro-Ordovician and Middle to Early Ordovician age. The Triassic and Late Permian components show a 20–30° cumulative counterclockwise offset with respect to the Indo-Pakistan apparent polar wander path (APWP). This indicates a post-Middle to -Late Triassic counterclockwise rotational movement of Kashmir over 65–75° with respect to peninsular Indo-Pakistan.

(D) A component of low inclination and predominantly east-west declination, which is observed only in rocks of pre-Panjal Traps age. A secondary origin associated with extrusion of the Panjal Traps flows is surmised, but could not be established beyond doubt. Alternatively, this component may be related to the regional foliation.

The Triassic and Late Permian primary component directions are predominantly of normal polarity. This suggests a pre-Punjabian (Late Permian) upper age limit for the Permo-Carboniferous (Kiaman) reversed polarity interval, which is earlier than the Early to Late Tatarian limit observed in the U.S.S.R.

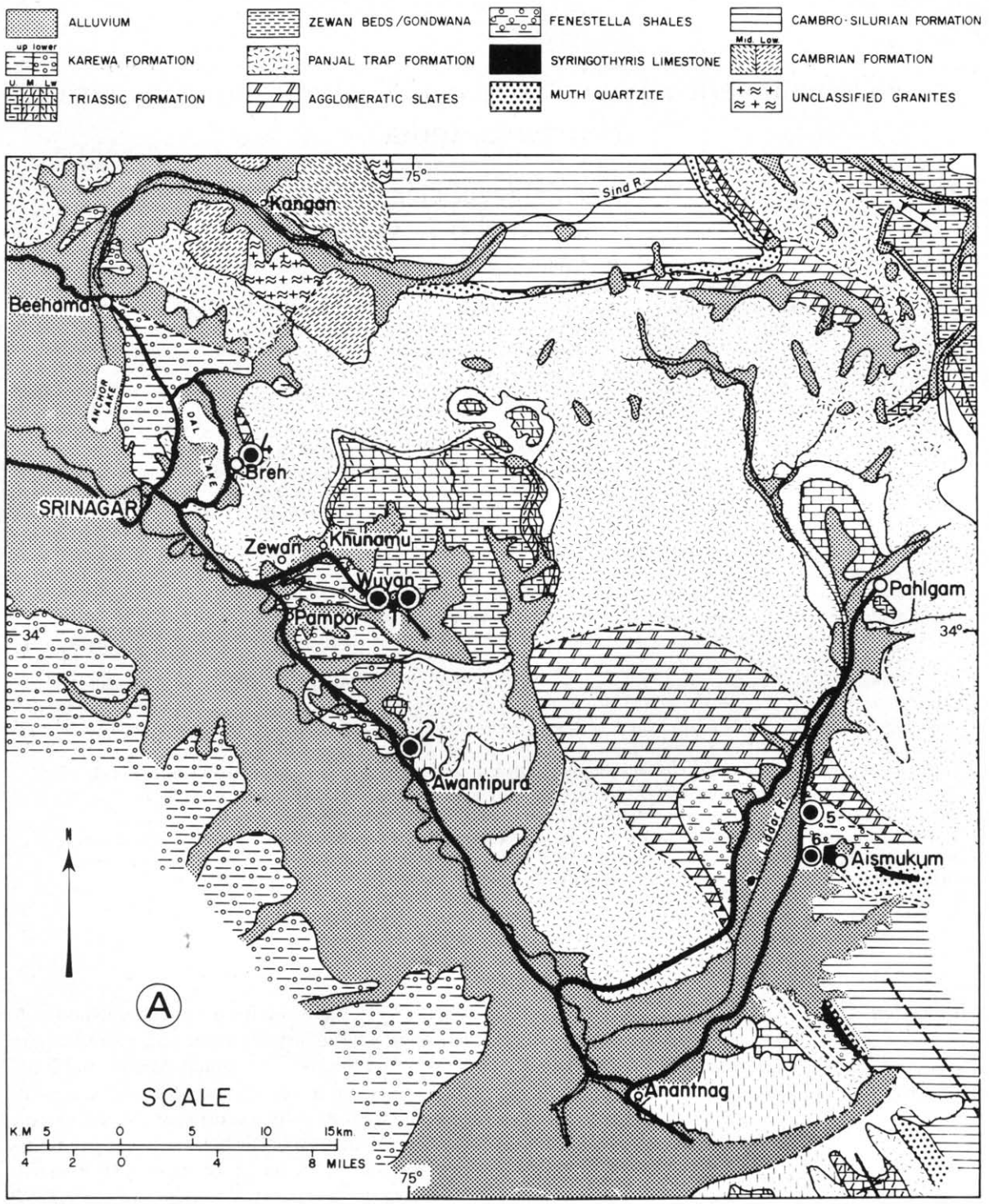
1. Introduction

Indian and foreign palaeomagnetic equipes have gathered over the last 20 years a considerable data base for Indo-Pakistan, from which a detailed Late Palaeozoic to Tertiary apparent polar wander path (APWP) [1,2] has been compiled. Middle and Early Palaeozoic data, in contrast, are lacking because of absence or restricted occurrence of Palaeozoic rocks on the peninsula. Detailing the largely undefined Palaeozoic segment of this APWP serves obvious goals. In wider perspective such an APWP segment may contribute to specify the movement

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ALLUVIUM
KAREWA FORMATION
TRIASSIC FORMATION
ZEWAN BEDS/GONDWANA
PANJAL TRAP FORMATION
AGGLOMERATIC SLATES
FENESTELLA SHALES
SYRINGOTHYRIS LIMESTONE
MUTH QUARTZITE
CAMBRO-SILURIAN FORMATION
CAMBRIAN FORMATION
UNCLASSIFIED GRANITES
75°
Sind R.
Kangan
Beehama
ANCHOR LAKE
DAL LAKE
Breh
SRINAGAR
Zewan
Khunamu
Wuyan
Pampor
Pahlgam
34°
Awantipura
Lidar R.
Aismukum
Anantnag
N
A
SCALE
KM 5 0 5 10 15km
4 2 0 4 8 MILES

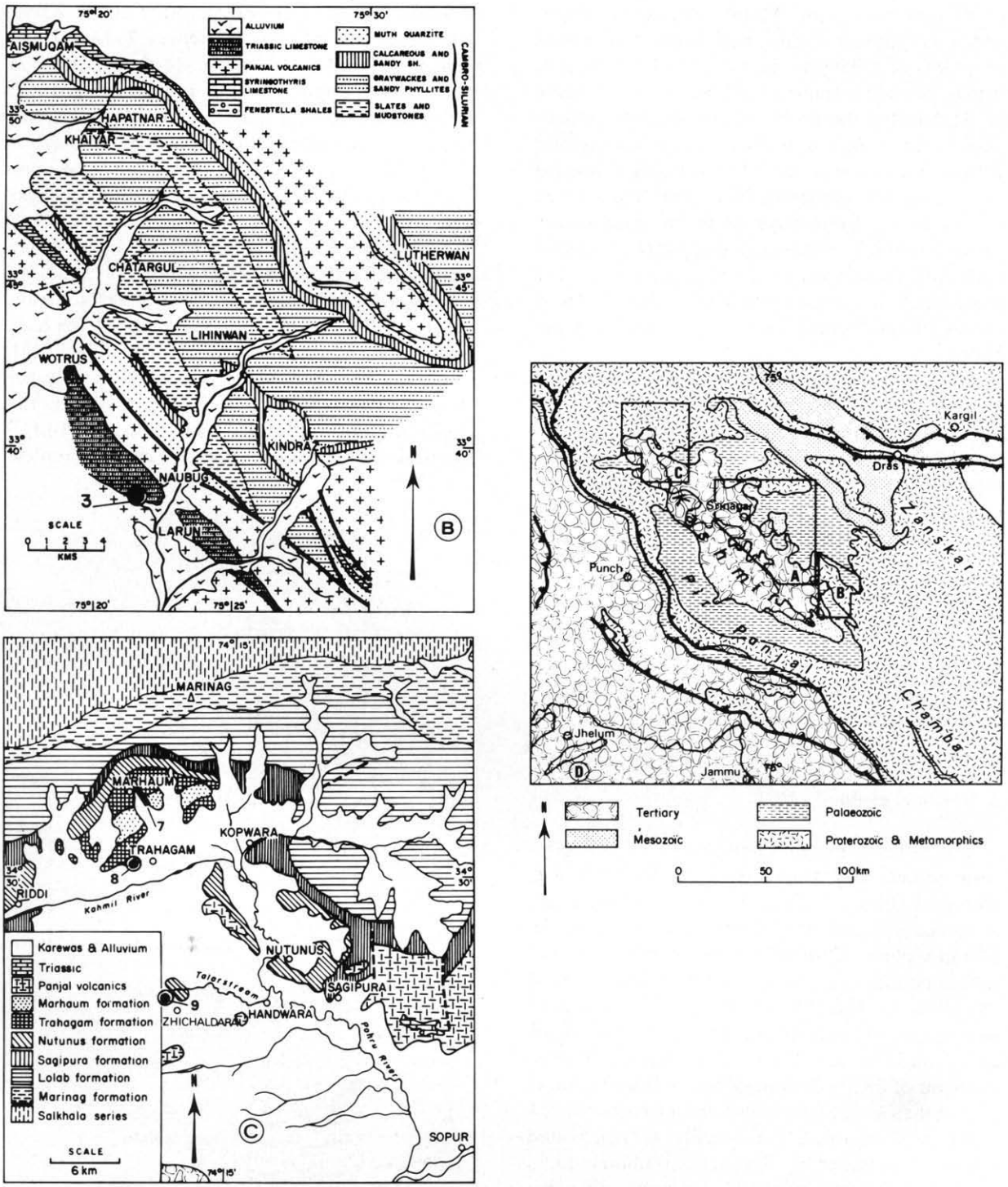


Fig. 1. Geological sketchmaps (A–D) of the Kashmir region after references 8, 10, 13, 41–44, with sampled localities indicated. See Table 1 for sampling details.

of Gondwana during Middle Palaeozoic times, which is not yet established within acceptable standards of reliability. In more local perspective, it may provide a frame of reference indispensable in establishing the probable pre-Mesozoic Gondwanic affinity of continental blocks now present within south central Asia. The Tethyan Himalaya with its nearly continuous and rather undisturbed Cambrian to Cretaceous platform sedimentary succession forms a challenging objective to obtain such data, although their direct interpretation and representation for peninsular Indo-Pakistan may not be straightforward because of tectonic complications.

The Tethyan sequence is situated north of the Main Central Thrust (MCT), which together with major thrusts in the Lesser Himalaya represents a zone of major intracontinental underthrusting. Rotational and less importantly so "translational" movements relative to peninsular Indo-Pakistan have to be detailed, before palaeomagnetic data from the Tethyan sequence can be used to detail the Indo-Pakistan APWP. This was possible in a study of the Tibetan Sedimentary Sequence of the Thakkhola region in Nepal [3], in which Mesozoic and Late Palaeozoic results were obtained. With a similar objective we studied Cambrian to Triassic sediments from Kashmir.

2. Regional geology

The rather complete Cambrian to Triassic platform sedimentary sequence of the Kashmir synclinorium (Figs. 1, 2) in the northwestern Himalaya is metamorphosed only locally and at low grade. Metamorphism is more intense in the mountain ranges bounding the Kashmir Valley south and to the north [4]. The southern Pir Panjal Range consists largely of Salkhala and Dogra metasediments of Precambrian age. The northern Zanskar Range, made up of highly deformed metasediments, forms part of the Great Crystalline Axis or Central Gneiss which was regarded traditionally as reactivated Precambrian basement. Recent fossil findings [5,6], however, prove the presence of younger Phanerozoic rocks.

For a detailed overview of the Palaeozoic-Mesozoic stratigraphy and structure of the Kashmir Valley we refer to references 7–13 for the Palaeozoic, and to reference 14 for a detailed study of the Upper Permian and Lower Triassic. We confine ourselves here to a summary description of major outlines of the regional geological setting. The stratigraphy of the sampled sections (Figs. 1, 2) is described in greater detail in the next section.

The northwestern part of the Kashmir Valley shows a well developed and fossiliferous Cambrian succession of argilite, ferruginous shale, subgraywacke and lenticular limestone, mapped for long as Cambro-Silurian but now subdivided [9,12,13]. The Ordovician-Silurian, in contrast, is poorly developed. A regional unconformity encompasses the Upper Silurian to Middle Carboniferous [7,10,11]. Southeastern Kashmir shows a more complete

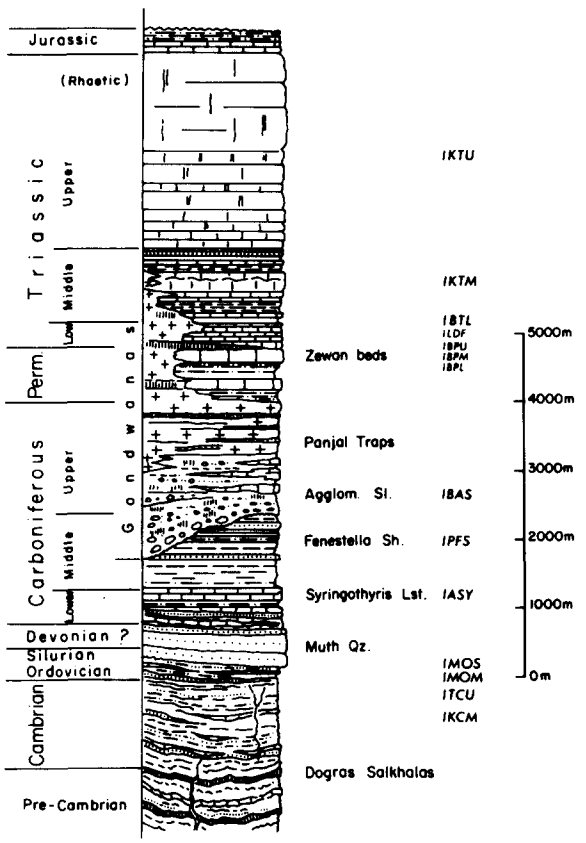


Fig. 2. Stratigraphical ideal column of Kashmir after Gansser [8].

Palaeozoic record, although Cambrian rocks have not been identified positively sofar. Volcanic activity was dominant during the Late Palaeozoic as evidenced by the Panjal Traps and the underlying Agglomeratic Slates. The origin of the latter has been disputed. Pyroclastic content indicates explosive volcanism prior to extrusion of the Panjal Traps, but intercalated tilloids and tillites point to a glacial origin. The Panjal Traps are made up of basic lava flows, up to 2500 m in thickness. Extrusion was most intense in northwestern Kashmir during the middle Early Permian, but diminished during the Middle Permian. Volcanic activity ceased by the Early Triassic [4], though flows maybe as young as Late Triassic occur in western Kashmir. According to Nakazawa et al. [14] this latter phase of volcanic activity postdates the Panjal Traps extrusion proper. The Upper Permian to Lower Triassic sedimentary succession was studied recently in great detail [14]. The Triassic and in particular the Upper Triassic, is very extensively developed. Apart from some local Jurassic occurrences no other Mesozoic outcrops are known. A large part of the basin is covered by nearly flatlying Plio-Pleistocene lacustrine sediments of the Karewas Formation [15].

The Palaeozoic-Mesozoic sequence of Kashmir and its basement of Salkhalas was transported southwards over the Panjal Thrust onto the well developed Tertiary molasse sequence of Jammu (Fig. 1). The contact zone is complicated by the Murree Thrust which outcrops slightly south of the Panjal Thrust (Fig. 1D) [11]. This thrust complex represents the westwards continuation of the MCT and the Main Boundary Thrust (MBT), which form widely separated entities in the more central Himalayan region, eastwards of Simla.

3. Sampling

An extensive collection of more than 800 core samples, mainly carbonates, was taken from Lower Palaeozoic and Upper Palaeozoic/Lower Mesozoic sediments of the Kashmir Basin (Figs. 1, 2). Samples were collected as stratigraphic profiles (Table 1), with generally one core taken per bed. All samples were oriented with a magnetic and a solar compass.

Lower Palaeozoic sediments were sampled in the Shamsh Abari syncline and in the Marhaum syncline of northwestern Kashmir (Fig. 1C, locations 7–9). Middle Cambrian dark-grey coloured limestone was sampled at Zhichaldara (Fig. 1C, location 9) in a succession of laminated limestone and slaty shale. Laminated limestone of late Late Cambrian age (Trahagaum Formation) was sampled at Trahagaum (Fig. 1C, location B), and recrystallized white to greyish limestone forming the Lower to Middle Ordovician upper part of the Trahagaum Formation was sampled in the Marhaum Nala (Fig. 1C, location 7) in the northern limb of the Marhaum syncline. A graywacke-shale succession of the Upper Ordovician/Silurian Marhaum Formation was sampled further southwards along the same nala in the core of the Marhaum syncline.

Lower and Middle Carboniferous rocks were sampled in the classical section of the Liddar Valley (Fig. 1A, locations 5, 6). Flaggy sandy limestone of the lower Lower Carboniferous (early Tournaisian) Syringothyris Limestone was sampled in the northeastern limb of the Liddar anticline, close to Aismukum (Fig. 1A, location 6). Further north along the Aismukum-Pahlgam road and in the same limb of the Liddar anticline (Fig. 1A, location 5) we sampled fossiliferous sandy shale of the Middle to possibly Upper Carboniferous Fenestella Shales [9,11]. A sandy to sandy shale succession at the top of the Upper Carboniferous to Lower Permian Agglomeratic Slates [14], directly below the massive Panjal Traps, was sampled in a quarry at Bren (Fig. 1A, location 4). The Panjal Traps were not sampled during the present reconnaissance since they were studied recently in some detail [16]. Extensive sampling was carried out in the Upper Permian Zewan Beds (Punjabian and Djulfian) and Lower Triassic beds at Barsu (Fig. 1A, location 2). Middle and Upper Triassic successions were sampled at Khrew (Fig. 1A, location 1). Three members of the Zewan Beds, described in detail by Nakazawa et al. [14], were sampled: i.e., well bedded fossiliferous limestone (member A), calcareous shale and micaceous sandy shale (member B), and an alternation of calcareous sandstone, shale and limestone (member E). Overlying well bedded limestone of the Lower

TABLE 1

Sampling details

Formation/ identification	Locality [a] (°N,°E)	Age	Lithology	Section length (m)	Number of samples (specimens)	Initial intensity (mA m^{-1})	Bedding	
							dip direction (°)	dip (°)
Upper Triassic IKTU	Khrew (1) 34.06, 75.04	Trl	mass. lst.	21.5 +24.2	75 (89)	0.03 – 200	77 –80	38–59
Middle Triassic IKTM	Khrew (1) 34.06, 75.04	Trm	nod. flas. lst	45.5 +5.5	81 (93)	0.3 –600	57– 87	28–37
Khunamuh Fm. IBTL	Barsu (2) 33.90, 75.02	Tre	sand. lst	17.5	49 (67)	0.04 – 10	255–265	49–56
Triassic s.l. ILDF	Naubug (3) 33.63, 75.39	Tr	lst.	10.5	50 (65)	0.3 – 0.9	350– 12	53–64
Zewan Beds, mb. E IBPU	Barsu (2) 33.90, 75.02	Pl	sand. lst.	28.5	41 (43)	0.4 – 10	262–268	51
Zewan Beds, mb. B. IBPM	Barsu (2) 33.90, 75.02	Pl	calc. sh.	11.5	43 (52)	0.2 – 0.4	257	55
Zewan Beds, mb. A IBPL	Barsu (2) 33.90, 75.02	Pl	sand. lst.	18	45 (70)	0.5 – 7	265	59
Agglomeratic Slates, IBAS	Bren (4) 34.12, 74.90	Cl-Pe	sand. sh.	21.5	80 (115)	0.3 – 2	11	29
Fenestella Shales, IPFS	Aismukum (5) 33.85, 75.25	Cm-l?	sand. sh.	36	90 (122)	0.08 – 10	10– 15	43
Synringothyris Limestone, IASY	Aismukum (6) 33.85, 75.25	Ce (Tourn.)	flag. sand. lst.	100	63 (111)	0.03 – 20	298–334	32–43
Marhaum Fm. IMOS	Marhaum Nala (7) 34.61, 74.25	Ol-S	grayw.	10	42 (63)	0.1 – 60	103–118; 4–17	355; 9
Trahagaum Fm. (top), IMOM	Marhaum Nala (7) 34.61, 74.25	Oe-m	recr. lst.	32	57 (78)	0.005– 2	130–140	47–50
Trahagaum Fm. ITCU	Trahagaum (8) 34.38, 74.16	Cl	lam. lst.	10.5	36 (64)	0.1 – 3	279–325	10–14
Middle Cambrian IKCM	Zhichaldara (9) 34.34, 74.22	Cm	lam. lst.	31	52 (99)	0.05 – 2	305–324	47

[a] See locality numbers in figure 1A-C.

TABLE 2

Summary of palaeomagnetic directions and pole positions

No. [a]	Formation	Mean specimen direction						South pole position					Blocking temperature (°C)	Bedding	
		D (°)	I (°)	k	α_{95} (°)	N	polarity (N/R)	lat. (°S)	long. (°E)	dp (°)	dm (°)	E_{95} (°)		dip direction (°)	dip (°)
(A) Recent field component [b]															
1	IKTU	356	+44	66.5	2	62	N	−81	99	1.5	2.5		< 250		
2	IKTM	357	+53.5	40	14.5	4	N	−87.5	165.5	14	20		< 250		
3	IBTL	3	+47.5	65	3.5	29	N	−84	48.5	3	4.5		< 250, also 450–630		
4	ILDF	355	+51	15.5	11.5	12	N	−85.5	142	10.5	15.5		< 250		
5	IBPU	354	+46	29.5	6.5	18	N	−81.5	115	5.5	8.5		< 250		
6	IBPM	343	+37.5	12	9	22	N	−70	128.5	6	10.5		< 250		
7	IBPL	358.5	+42.5	42	5.5	18	N	−80.5	83.5	4	7		< 250		
8	IPFS	359.5	+45.5	21.5	4	62	N	−83	79	3	5		< 250		
9	IASY	359	+38.5	22	8.5	15	N	−78	79.5	6	10		< 200		
10	IMOS	4.5	+59.5	33.5	4	41	N	−83	105	4.5	6		< 250, also 250–600		
11	ITCU	1	+56	20	8.5	16	N	−88	96.5	9	12		< 250		
12	IKCM	352	+52	42	4.5	23	N	−83	152	4	6		< 200		
	Mean	356.5	+48	107	4	12		−83	106			3			
(B) Secondary, Middle to Late Tertiary component, not corrected for bedding															
13	IKTU	194.5	−46.5	24	4.5	48	pred. R3	−76	8	3.5	6		230–350, occ. 550		
14	IKTM	191	−23	12	9.5	21	R3	−66	48	5.5	10		100–500		
15	IKTM	25.5	+27.5	7.5	13	19	N2	−60	19	7.5	14		100–600		
16	IBTL	213.5	−30	12	8.5	26	pred. R1	−55	7	5	9.5		230–470		
17	ILDF	27.5	+40.4	15	24.5	4	N2	−63.5	2	18	29.5		200–300, occ. 550		
18	ILDF	193.5	−38.5	14	14.5	9	R3	−73	27	10	17		200–300, occ. 550		
19	IBPU	203	−13.5	125	11.5	14	R1	−55.5	31.5	6	11.5		260–380, occ. 500		

TABLE 2 (continued)

No. [a]	Formation	Mean specimen direction						South pole position					Blocking temperature (°C)	Bedding	
		D (°)	I (°)	k	α_{95} (°)	N	polarity (N/R)	lat. (°S)	long. (°E)	dp (°)	dm (°)	E_{95} (°)		dip direction (°)	dip (°)
20	IBPM	208	−37	11	17.5	8	pred. R1	−62	6	12	20.5		260–380		
21	IBPL	205.5	−30.5	29	5	27	R1	−61	16	3	5.5		260–560		
22	IBPL	208	−24.5	50	4	25	R1	−57	18	2.5	4.5		260–560		
23	IBPL	192	−37.5	51	4	26	R3	−73.5	32.5	3	4.5		260–560		
24	IBAS	187.5	−33	44.5	2	100	R3	−72.5	50.5	1	2.5		260–350		
25	IPFS	195	−51	24	3.5	66	R3	−77	171	3	4.5		250–350		
26	IASY	196.5	−37.5	44	2	91	R3	−70.5	22.5	1.5	2.5		230–300, occ. 380		
27	IMOS	159	−13	15	13	10	R	−56	113.5	7	13		250–350		
28	IMOS	331	+14.5	29.5	4.5	39	N	−52	125.5	2.5	4.5		250–350		
29	ITCU	24.5	+35	157.5	1.5	64	N2?	−63.5	13	1	1.5		230–300		
30	IKCM	155.5	−38	29.5	3	79	R	−65	139.5	2	3.5		230–300		
31	IKCM	18.5	+39.5	14.5	55	48	N2?	−70	15.5	4	6.5		230–300		
32	Mn. R3	192.5	−38	75	7	7		−76	38			6			
33	Mn. N2	24	35.5	149	7.5	4		−64.5	12.5			6			
34	Mn. R1	207.5	−27	73.5	9	5		−58.5	16			9			
(C) Primary components															
35	IKTU	278	−58	10	10	24	(21/3)	+15	128	11	14.5		300–530	77– 80	38–52
36	IKTM	288	−58	12	8	29	(12/17)	+8.5	123.5	8.5	12		300–550	57– 87	28–37

37	IBTL	267.5	−68.5	7.5	11.5	25	(22/3)	+27.5	119	16.5	19.5	270–500	255–265	49–65
38	ILDF	258	−64	7.5	13.5	18	(12/6)	+31	128.5	17	21.5	110–350 and 230–390, occ. 500	350– 12	53–64
39	IBPU	269	−52	6	11.5	32	N	+18	137.5	11	15.5	260, occ. 300–500	262–268	51
40	IBPM	262.5	−68	8.5	10.5	26	(23/3)	+30	121	15	17.5	260, occ. 300–500	257	55
41	IBPL	249.5	−65	8	11	25	(23/2)	+37.5	128.5	14.5	17.5	400–565	265	59
42	IASY	197	+44	8	13	18	(2/16)	+28.5	58	10	16.5	300–500	298–334	32–43
43	IMOS	258.5	+55.5	12.5	10	18	R	+11.5	20	10	14.5	350–600	103–118; 4–17	355; 9
44	IMOM	346.5	+55	22.5	4	53	(9/44)	+79	353	4	5.5	350–500	130–140	47–50
(D) Component of uncertain interpretation (north or south pole indicated)														
45	IBAS	274.5	+22.5	9.5	13.5	14	(10/4)	−10	172	7.5	14.5	< 300	11	29
46	IPFS	261.5	+11	5.5	13.5	25	(15/10)	+4	174.5	7	13.5	250–550	10– 15	43
47	IASY	262.5	−20.5	8	11	24	(17/7)	+12	160.5	6	11.5	200–350, 300–500	298–334	32–43
48	IMOS	110.5	+42	8	11	24	N	−2	133	8.5	13.5	250–500	103–118; 4–17	355; 9
49	IMOM	290.3	−27.5	11	15	10	(4/6)	−7.5	140.5	9	16.5	350–500	130–140	47–50
50	ITCU	256	−9	5	14	25	(17/8)	+14	168.5	7	14	300–500	279–325	10–14
51	IKCM	109.5	+22.5	5	21	13	R	−9	143.5	12	22.5	250–500	305–324	47

[a] Numbers correspond with identifications in Figs. 5, 6B.
[b] Present field direction in Kashmir: $D = 358.5°$, $I = +51°$.

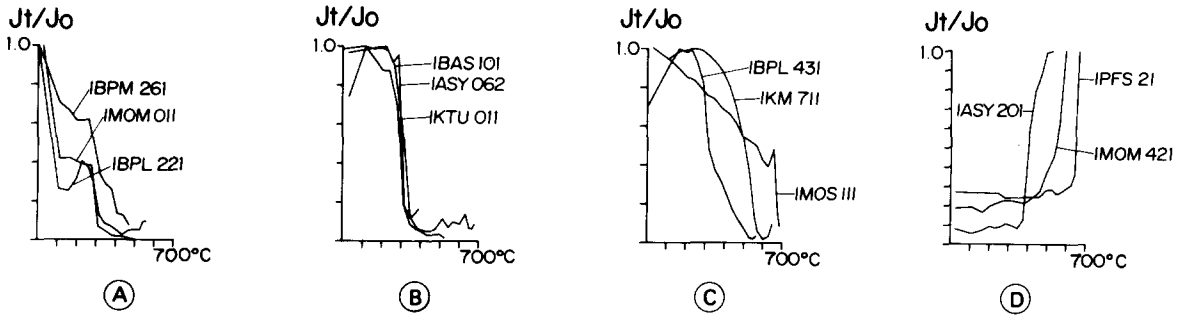
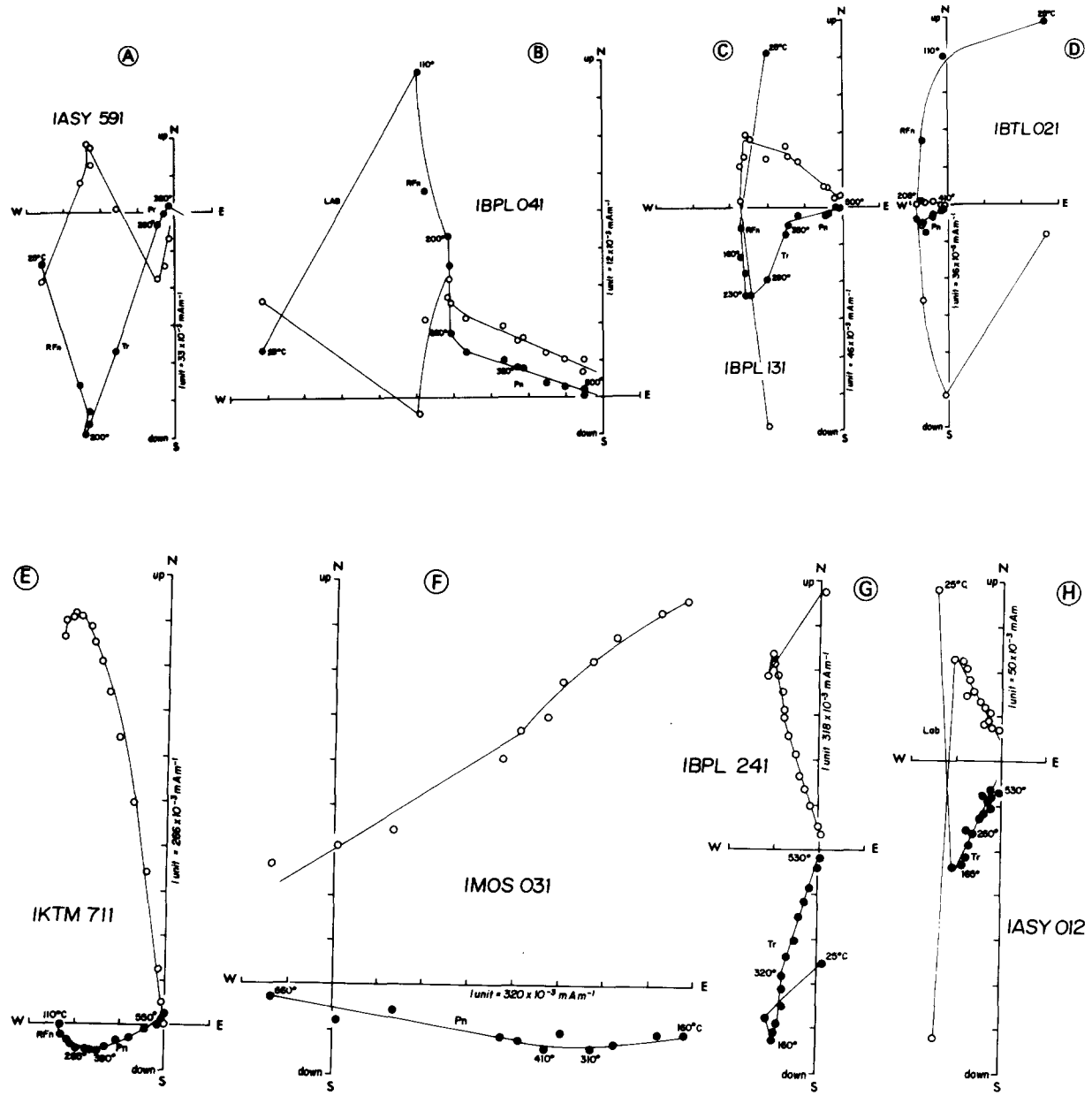


Fig. 3. Representative normalized curves showing the decay of remanent magnetization (A, B, C) and changes in magnetic susceptibility (D) during thermal treatment.



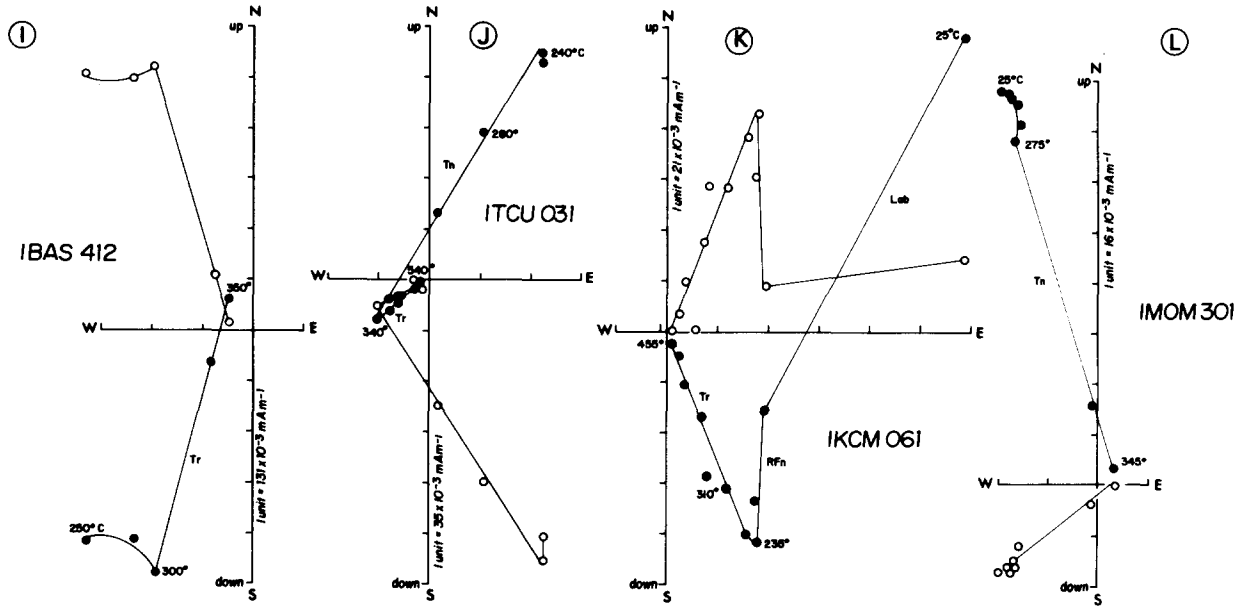


Fig. 4. Demagnetization diagrams of some representative specimens. The points denote successive positions—in orthogonal projection—of the end points of the resultant magnetization vector during progressive thermal demagnetization. Open circles denote projections on the vertical east-west plane, dots projections on the horizontal plane. Numbers denote successive peak temperature values. RF = recent field component, T = Middle to Late Tertiary secondary component, P = primary component; n = normal polarity, r = reversed polarity. Results are not corrected for bedding.

Triassic Khunamuh Formation (member G) was sampled in the same profile. Siliceous nodular and red flaggy limestone of late Early to Middle Triassic age was sampled in a hill-top section near the temple at Khrew. Blue-grey massive and brittle limestone of the very thick Upper Triassic succession was sampled at a short distance further eastwards (Fig. 1A, location 1) in two short sections about 1000 m stratigraphically apart. Another section of well bedded limestone of unspecified Triassic age was sampled directly south of Naubug (Fig. 1B, location 3) at the base of a limestone succession directly overlying the Panjal Traps.

4. Analysis and results

Sample preparation, progressive thermal demagnetization (10–15 steps) in feedback controlled furnaces [17], and measurement on a two-axis ScT cryogenic magnetometer followed routines described in detail elsewhere [3,18]. Directional analysis was based on the specimen level. Individual Zijderveld [19,20] plots were visually analyzed for straight-line segments and directions of individual components were computed with an adapted version of Kirschvink's [21] principal component analysis program. Interpretation of the data was based on density distribution figures.

Three well interpretable magnetic components could be distinguished throughout the Palaeozoic and Mesozoic rocks, with a fourth component of uncertain origin being observed in Middle and Lower Palaeozoic rocks only. The common magnetic characteristics digested from the large amount of data gathered (804 samples, 1131 specimens) are described hereunder, with a more detailed account on the characteristics of individual rock sequences listed in Tables 1 and 2.

(1) A recent field component of normal polarity and of considerable intensity (10–80% of initial natural remanent magnetization (NRM)). This component was removed below 200° to 250°C (Figs. 3A, 4A–E, 5F).

(2) A secondary component of Middle to Late Tertiary origin, and of predominantly reversed polarity which was removed in a restricted temperature interval between 250° and 300°C (Figs. 3A,B, 4A,C,G–L, 5B,C). This component represented up to 40% of initial NRM in the Permian and Triassic carbonates and up to 80% in the Middle and Lower Palaeozoic rocks.

(3) A primary magnetic component of rather low intensity, which became removed gradually between 300° and 550°C (Figs. 3C, 4A–F, 5E). This component could well be identified in the Permian and Triassic rocks, but less so in the Middle and Lower Palaeozoic formations. The Permian and Triassic carbonates showed predominantly normal polarities. The Syringothyris Lime-

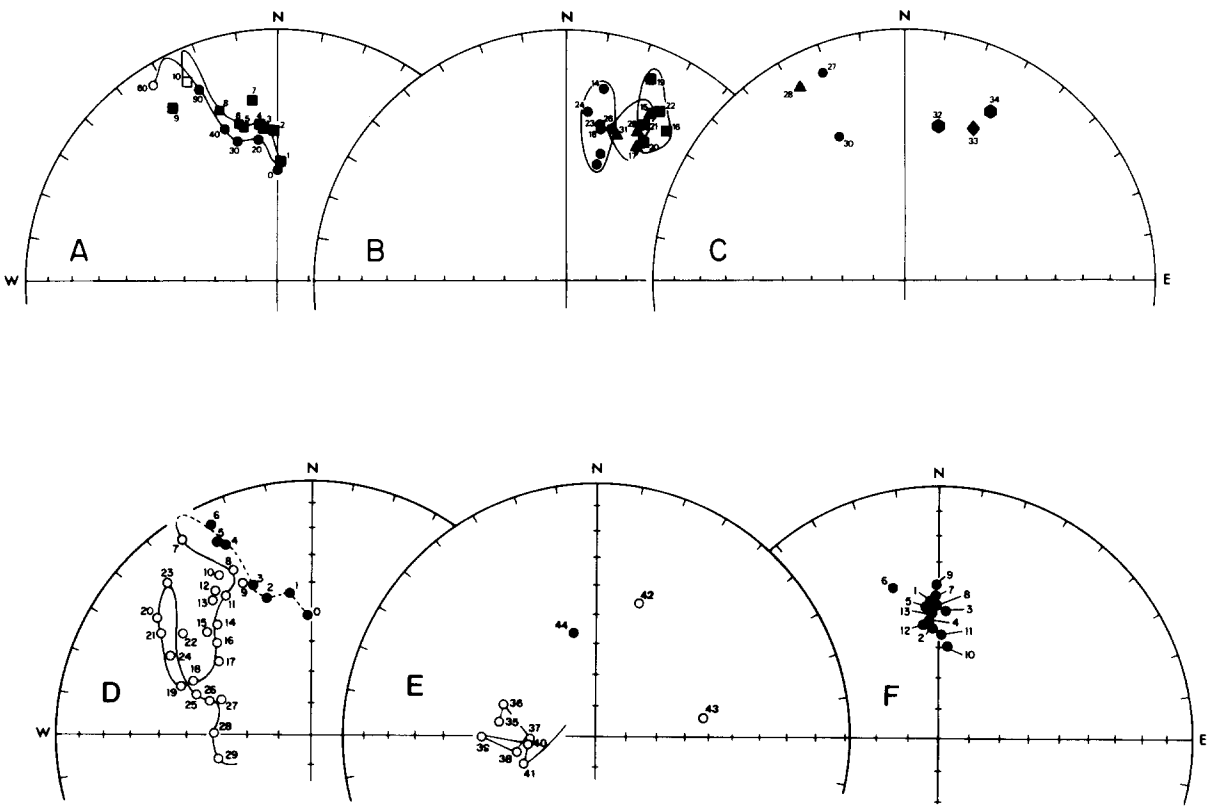


Fig. 5. A. Directions expected in the Kashmir region according to Tertiary palaeomagnetic data from the Indian plate (circles: ages in millions of years, see Table 3), and according to Australian Tertiary data rotated to the Indian plate (squares; see Klootwijk and Peirce [28, Table 1]): *1* = Newer Volcanics, Victoria (0–5 Myr), *2* = Nandewar Volcano (17–18 Myr), *3* = lavas and dykes, southeast Queensland (20–22 Myr), *4* = Main Range Volcanics (20–22 Myr), *5* = Canaway profile (Upper Oligocene?), *6* = Tweed Volcano (20–22 Myr), *7* = Liverpool Volcano (33.7 Myr), *8* = Barrington Volcano (51.7 Myr), *9* = Older Volcanics, Victoria (Palaeocene/Eocene?), *10* = Mornay profile (Maastrichtian/Lower Eocene). B. Secondary magnetization directions of Middle to Late Tertiary age observed in central and eastern Kashmir, not corrected for bedding. Dots (triangles) indicate reversed (normal) polarity directions. See Table 2. C. As above, for northwestern Kashmir (Table 2, *27, 28, 30*). Mean component directions for the three groupings shown in Fig. 5B are indicated by symbols *32–34*. D. Expected directions in the Kashmir region for Late Palaeozoic to Tertiary times according to palaeomagnetic results from peninsular Indo-Pakistan and DSDP cores of the Indian plate (see Table 3). 0 = present axial geocentric dipole field direction. E. Primary magnetization directions for the Palaeozoic/Lower Mesozoic sequence of Kashmir. See Table 2. F. Locality mean directions of recent field components (circles). Compare with the present axial geocentric dipole field direction in Fig. 5D and Table 2. All figures are in equal area projection. Full (open) symbols denote downward (upward) pointing directions.

stone, the Marhaum Formation and the top of the Trahagaum Formation, in contrast, showed nearly exclusively reversed polarities (Table 2).

(4) Another component of predominantly east-west declination and rather shallow inclination, was identified only in Middle and Lower Palaeozoic formations of pre-Panjal Traps age. This component showed mixed polarity directions and was broken down gradually between 200° and 300°C or between 300° and 500°C.

Susceptibility increased in general at an excessive rate after successive heatings above 400–500°C (Fig. 3D). A concomittant excessive increase in a laboratory-induced VRM impeded remanence measurements after heatings at higher temperatures.

5. Interpretation

Interpretation of the origin and polarity of the various components listed above was based on comparison of their directions and pole positions with the Indo-Pakistan APWP and the general palaeomagnetic data from Gondwana. Continuous care was taken in this analysis to account for the regional geological setting and in particular for possible rotational movements of thrust units relative to the Indian shield. The Indo-Pakistan APWP, at least for post-Late Palaeozoic time, is now sufficiently well established [22] and shows sufficient detail to avoid the danger of circular reasoning in such an interpretation. Our sampling was not directed to extensive application of the fold test, as this test is of limited conclusive value in regions of recent tectonic activity like Kashmir (up to Plio-Pleistocene time).

The recent field component (1) was clearly identifiable from the individual diagrams and its interpretation needs no further discussion. Existence of a secondary component of Tertiary origin (2) was expected on basis of its general presence in extrapeninsular Indo-Pakistan [1,2,23,24], and was identified positively after accounting for rotational movement of thrust units. The results fit well with the regional tectonic setting and with palaeomagnetic evidence for similar rotations of other thrust units in the northwestern Lower Himalaya [23,24].

The interpretation of a primary origin for component (3) invokes a more complicated tectonic history, but is beyond any reasonable doubt for at least the internally consistent and abundant Upper Permian and Triassic data.

The origin of component (4) is less clear. A speculative and complex origin, which is a corollary of the conclusions reached for components (2) and (3), will be discussed.

Formation mean directions for component (2) form a loose grouping in the first quadrant (Fig. 5B). Correction for bedding disperses this grouping, which indicates a post-tectonic origin. These component directions are predominantly of reversed polarity with several normal polarity directions notably confined to the centre of the rather loose grouping. This suggests that the population may be made up of separate reversed-normal-reversed subgroups (Table 2, Fig. 5B,C). Comparison of the loci of the mean pole positions for these three subgroups (which lie on a small circle around the sampled site) with the Indo-Pakistan APWP (Fig. 6B) indicates a Middle to Late Tertiary secondary origin for this component. Clearly, this component relates to the Oligo-Miocene closure phase of isotopic systems observed throughout the Himalayan range [4,25]. Further comparison of directions and pole positions (Figs. 5A–C, 6B) indicates a clockwise rotation of the Panjal Nappe with respect to the Indian shield over approximately 45° since acquisition of the older reversed component. Such a rotation is in accord with an emerging picture of clockwise rotations of thrust units in the wider Western Himalaya region [23,24]. Recently obtained data from Oligo-Miocene red beds of the Murree Formation (Klootwijk et al., in preparation) in the southwards adjacent Riasi Thrust sheet show similarly displaced pole positions (Fig. 6B, poles C–E), which adds convincing support for the interpretation of a clockwise rotation of the Panjal Nappe.

Several of the secondary component directions observed in northwestern Kashmir and in particular those observed in the northwestern limb of the Marhaum syncline correspond with results for Indo-Pakistan (Figs. 5A,C, 6B), and show no indication for such a clockwise rotation. This leads to the conclusion that the area studied in northwest-

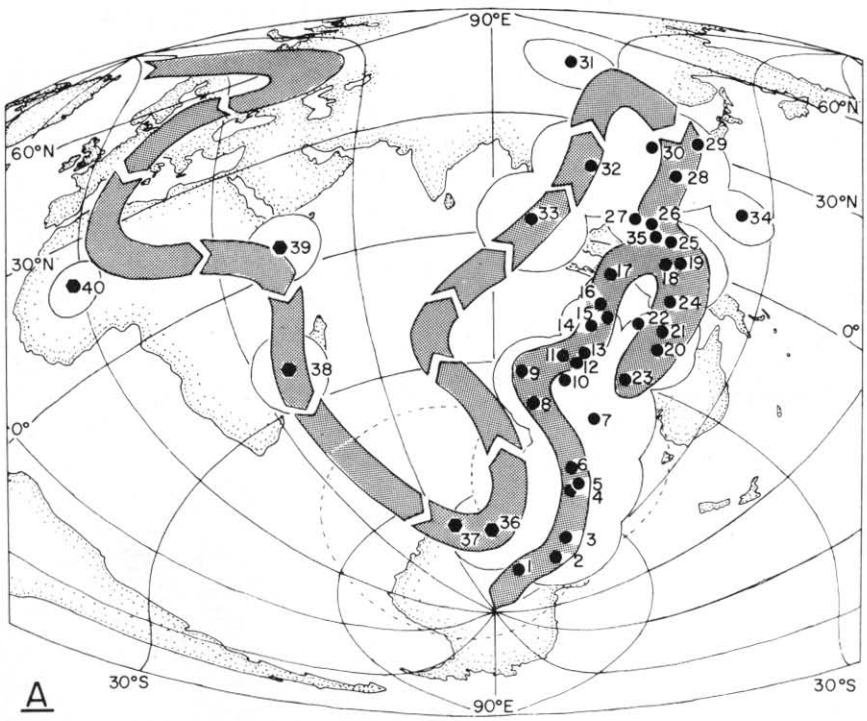
TABLE 3

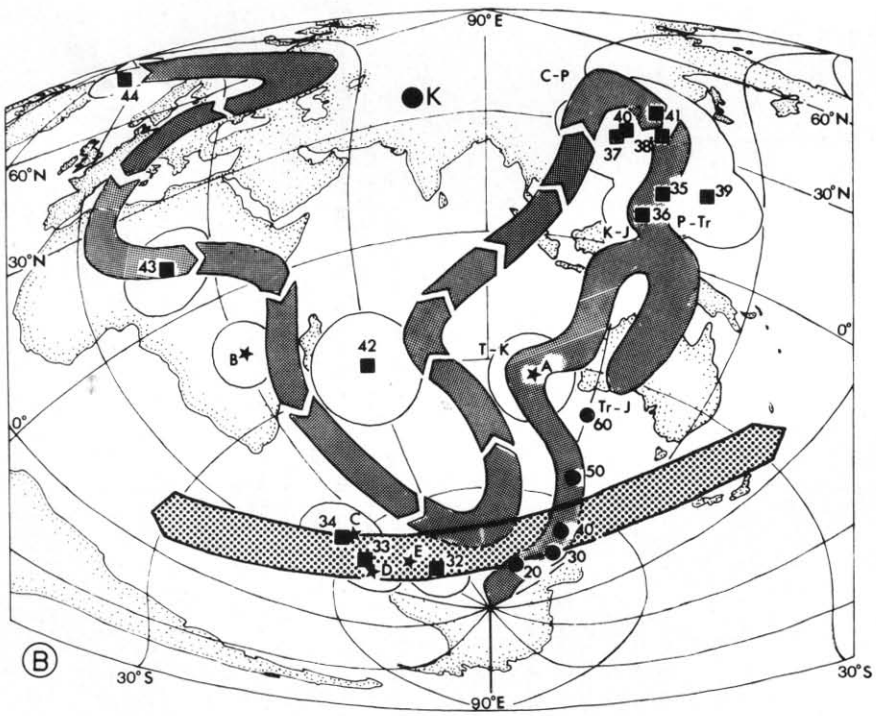
Pole positions for Indo-Pakistan

No.	Rock unit	Sampling area (°N,°E)	Age (Myr)	Direction						South pole position				
				D (°)	I (°)	k	α_{95}(°)	N [a]		lat. (°S)	long. (°E)	dp (°)	dm (°)	E_{95} (°)
1	DSDP cores	Indian Ocean	5–22.5 (20)					2		−78.5	114.5			
2	DSDP cores	Indian Ocean	25–35 (30)					4		−72.5	136			3.5
3	DSDP cores	Indian Ocean	35–45 (40)					3		−67.5	132			12
4	DSDP cores	Indian Ocean	45–55 (50)					4		−57.5	121.5			9
5	Sanjawi Lst. [b]	30.2, 68.2	Tpa-e	332	+15	13.5	14.5		(67)	−55.5	123.5	7.5	15	
6	Brewery Lst.	30.1, 66.6	Tpa	331.5	+8.5	8.5	22		(7)	−52.5	118.5	11	22	
7	DSDP cores	Indian Ocean	55–65 (60)					5		−40.5	120			10.5
8	Deccan Traps upper N epoch		60–65					32		−39.5	101.5			5.5
9	Deccan Traps lower R epoch		60–65					47		−32.5	97.5			3
10	Tirupati Sst.	16.8, 81.2	Ke	332	−52.5	52.5	11		(43)	−33.5	109.5	10.5	15	
11	Tirupati Sst.	16.8, 81.2	Ke	330	−58	443	4	4	(65)	−28	107.5	4.5	6	
12	DSDP cores	Indian Ocean	65–75 (70)					2		−29	111.5			
13	Satyavedu beds	13.5, 79.9	Ke	321	−58	80	4		(13)	−26.5	113	4.5	6	
14	Sylhet Traps	25.0, 91.0	Ke??	332	−59		7		(25)	−20	113.5	8	10.5	
15	Goru Fm.-Parh Lst. [b]	30.2, 68.2	Ke-l	309.5	−40.5	32.5	3		(76)	−17.5	117	2	3.5	
16	Goru Fm-Parh Lst.	30.1, 66.6	Ke-l	307	−43.5	16.5	4		(86)	−14.5	115	3	5	
17	Rajmahal Traps	24.5, 87.6	100–105	316	−65	85	2.5	48	(294)	−7.5	116.5	3	3.5	
18	Loralai Lst. [b]	30.2, 68.2	Jm	228	−43	9	8		(41)	−2	128	6	10	
19	Chiltan Lst.	30.1, 66.6	Jm-l	104	+38	17	105		(30)	−0.5	131.5	7.5	12.5	
20	Australia [c]		Jm					3		−21.5	130.5			9

21	Australia [c]		Je					2		−17	130.5			
22	Antarctica [c]		Jm					5		−17	124.5			14.5
23	Parsora beds	23.5, 81.5	Trl	320	−39	200	5.5	3	(48)	−30	125	4	7	
24	Pachmarhi beds	22.8, 78.8	Trl	117	+49	33	4.5	3	(31)	−10	130.5	4.5	6	
25	Kamthi beds	20.0, 79.0	Pl-Tre	101.5	+58.5		6.5	22	(105)	+4	128.5	7	9.5	
26	Mangli beds	20.5, 79.0	Pl-Tre	100	+63	43.5	4.5	2	(23)	+7	124.5	5.5	7	
27	Panchet beds	23.9, 85.2	Pl-Tre	110.5	+69	49	6		(13)	+7	120.5	8.5	10	
28	Kamthi beds	20.0, 79.0	Pl-Tre	79.5	+61.5	63	2.5	6	(57)	+21	129.5	3	3.5	
29	Talchir beds	21.5, 78.9	P-C	66	+59	109.5	3		(23)	+31.5	134.5	3.5	4.5	
30	Krol-A Lst. [d]	30.2, 78.4	P-Tr	306.5	−67.5	8	15		(14)	+26.5	123.5	21	25	
31	Blaini Lst. [d]	30.2, 78.4	P-C	269.5	−74.5	7	7.5		(62)	+46.5	109	12.5	13.5	
32	Alozai Fm.	30.1, 66.5	P-C?	267.5	+66	6	12		(28)	+18	111	16	20	
33	Lower Blaini Diamictite [d]	30.2, 78.4	P-C	354	−72	7	12		(23)	+3	98.5	12.5	19	
34	Warchha Sst. [e]	32.6, 72.5	Pe	85.5	42.5	113	6.5	6	(61)	+16.5	143.5	5	8	
35	Warchha Sst. [e]	32.7, 72.7	Pe	108.5	53.5	278	7.5	3	(25)	+4.5	124.5	7	10.5	
36	S.E. Australia [f,g]		D				9.5			−69.5	88.5	5	9.5	
37	S.E. Australia [g]		D				28.5			−67.5	67.5	16.5	30.5	
38	S.E. Australia [g]		Sm-l				9			−22	38.5	4.5	9.5	
39	S.E. Australia [g]		Se-l				8.5			+5.5	44	4	8.5	
40	S.E. Australia [g]		Om-l				6			+19	358.5	6	8.5	

[a] Number of sites (number of samples between parentheses).
[b] Corrected for a 50° clockwise rotation of the Loralai Range [18].
[c] Mean pole transferred to the Indian plate according to seafloor spreading data [18,28].
[d] Results corrected for a 45° clockwise rotation [23].
[e] Results not corrected for a possible rotation of the Salt Range [29].
[f] Interpreted as a secondary component [26,27].
[g] Results from southeastern Australia [26,27] transferred to the Indian plate (see footnote c).


90°E
60°N
30°N
0°
30°S
90°E
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90°E
60°N
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30°S
90°E
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ern Kashmir may well have lagged behind as it is situated near to the supposed pivot point of rotation in the Western Himalayan Syntaxis. Such a change in secondary magnetic directions coincides with a marked regional change in the alignment of fold axes. These show a NW-SE trend in central and eastern Kashmir, but are east-west aligned in northwestern Kashmir, reflecting the influence on local tectonics of the nearby syntaxis.

Pole positions for component (3) of the Triassic and Upper Permian sequence in central and eastern Kashmir form an internally consistent group, resembling the Permo-Triassic segment of the Indo-Pakistan APWP but displaced counterclockwise over 20–30° (Figs. 5D,E, 6A,B). Such a resemblance is duly expected as the Himalayan region forms the northern margin of the Indo-Pakistan subcontinent, reactivated during the Alpine orogeny. This consistent counterclockwise offset points to a post-Middle to -Late Triassic rotational movement of the Kashmir region in a sense opposite to the Tertiary clockwise rotation. The magnitude of the former rotation can be estimated at 65–75° (45° + 20 to 30°).

The Carboniferous to Cambrian segment of Indo-Pakistan's APWP is undefined because of lack of suitable outcrops on the peninsula. This hindered interpretation of the results for component (3) of the Lower and Middle Palaeozoic formations sampled in northwestern Kashmir, the more so as data for that period are available only scarcely from other Gondwana continents. A notable exception, however, forms a recently obtained [26,27] palaeomagnetic record of a Lower to Middle Palaeozoic succession from the Tasman fold belt of southeastern Australia. It has not been established firmly as yet whether these data can be taken as representative for the Australian continental block and for Gondwana. If so, they may serve as a frame of reference for interpretation of the present results from Kashmir. These Ordovician to Devonian pole positions from Australia have been rotated to Indo-Pakistan [28] following reconstruction parameters for the Gondwana configuration of Smith and Hallam [29], adapted for the Australia-Antarctica fit according to Griffiths [30] and Norton and Molnar [31]. A tentative extrapolation of the Indo-Pakistan APWP based on these Middle Palaeozoic results and on recently obtained Cambrian results from Australia [32–34] is shown in Fig. 6A. Pole positions for the Lower to Middle Ordovician top of the Trahagaum Formation and for the Siluro-Ordovician Marhaum Formation (Fig. 6B, *43*, *44*) are in broad agreement with this tentative APWP. Both sets of data, however, bear insufficient detail to show conclusive evidence for the 20–30° cumulative counterclockwise rotation observed for the Late Permian and Triassic results. This reduces the potential of the present data to extrapolate the Indo-Pakistan APWP backwards into the Palaeozoic. The pole position for the Early Carboniferous Syringothyris Limestone (Fig. 6B, *42*) is not in complete accord with this tentative APWP, though it is not far off a preliminary pole position for the Carboniferous Phe Volcanics (Fig. 6B, *33*) from the Zanskar Range. Apart from a possible inappropriate extrapolation of the southeast Australian data, we suggest as alternative explanations for this offsided Late Palaeozoic pole position, either that we have mistakenly interpreted this palaeomagnetic result to be of primary origin, or that the Carboniferous segment of the Indo-Pakistan APWP may be more complex than sketched.

Fig. 6. A. Phanerozoic APWP for Indo-Pakistan, based on (south) pole positions from peninsular and extrapeninsular Indo-Pakistan (Table 3, poles *1 – 35*) and transferred pole positions from southeast Australia ([26–28, 32–34]; Table 3, poles *36 – 40*). B. Comparison of the Indo-Pakistan APWP as shown in Fig. 6A with primary (Table 2, *35 – 44*) and secondary (Table 2, *32 – 34*) mean pole positions from Kashmir. The pole positions for the Phe Volcanics (Carboniferous: 14.5°S,30.5°E; $dp = 6°$, $dm = 7.5°$ [35]) and for the Panjal Traps (Lower Permian: 33.5°S,102°E; $dp = 6.5°$, $dm = 11°$ [17]) are shown for comparison. The loci of the secondary mean pole positions intersect the Indo-Pakistan APWP in its Middle to Late Tertiary segment. Their offset indicates a clockwise rotation of the sampled region over approximately 45° with respect to peninsular Indo-Pakistan. Pole positions from the Oligo-Miocene Murree Formation of the Riasi thrust sheet which is southwards adjacent to the Panjal Nappe indicate a similar rotation (*C*: 59°S,22°E; $dp = 1.9°$, $dm = 3.6°$. *D*: 66.5°S,6°E; $dp = 3.2°$, $dm = 5.3°$. *E*: 71°S,23°E, $dp = 3.5°$, $dm = 6°$). Aitoff world projection. Ellipses of confidence (95%) are shown outside the shaded APWP trajectory only.

The fourth magnetic component has been observed exclusively in rocks predating the Permian to Triassic extrusion of the Panjal Traps. This suggests a secondary origin associated with extrusion of these flows. This magnetic component of both normal and reserved polarity, shows after correction for bedding shallow inclinations in the second and fourth quadrant, but with a predominant east-west declination. Corresponding (south?) pole positions form two groupings east of the Indo-Pakistan APWP (Fig. 7), far off the Panjal Trap pole which is possibly of a secondary Early Tertiary origin [16]. If of Panjal Trap origin indeed, unfolding for present-day bedding inappropriately includes a correction for an earlier phase of folding over ENE-WSW folding axes [7,13]. For reason of a rather wide regional hiatus [7,10,11] this phase cannot be dated better than Middle Palaeozoic. The resulting uncertainty in pole positions is indicated by the double-headed arrows in Fig. 7.

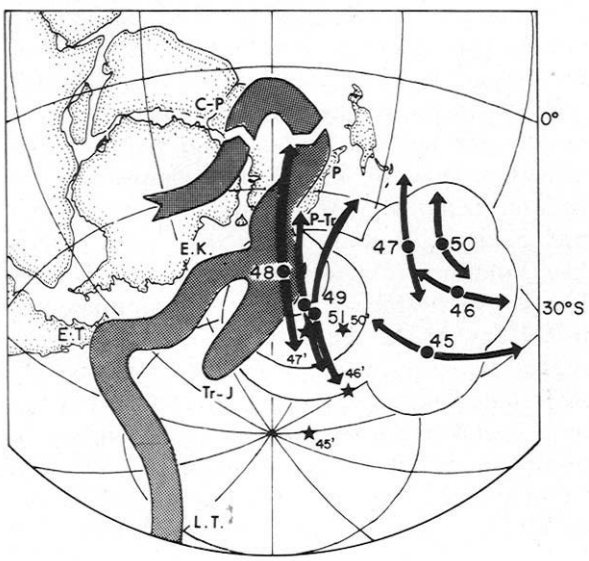


Fig. 7. Pole positions for magnetic component (D) (Table 2, *45–51*) which are off the Indo-Pakistan APWP and whose origin is uncertain. The double-headed arrows indicate changes in pole positions resulting from a possibly redundant unfolding for a Middle Palaeozoic folding phase over ENE-WSW axes, which can be argued for on basis of a suggested secondary Panjal Traps related origin of the magnetic component. Pole positions *45′–47′* and *50′* have been corrected for the effect of a cumulative counterclockwise rotation over 30°. Gondwana continents reassembled in the Smith and Hallam reconstruction [29–31], with all pole positions rotated to Africa in its present position. See text.

The pole positions for this fourth component concentrate into two groups, which notably are also regionally confined to either northwestern Kashmir (poles *48, 49, 51*) or southeastern Kashmir (poles *45–47*, with pole *50* from northwestern Kashmir as exception). That is, confined to regions for which the secondary Middle to Late Tertiary component [2] indicates either presence (southeastern) or absence (parts of northwestern Kashmir) of the 45° clockwise rotation. The poles from the northwestern Kashmir group position close to the Middle Triassic segment of the Indo-Pakistan APWP. Consequently a secondary Panjal Traps related origin could be argued for, as the long-lasting igneous activity had ceased by that time. The poles from southeastern Kashmir, in contrast, position off this APWP, but can be brought into better agreement with those from northwestern Kashmir after correction for the 20–30° cumulative counterclockwise rotation (Fig. 7). The arguments given above invoke a post-Middle to -Late Triassic age for the phase of counterclockwise rotation over 65–75°, which affected central and southeastern Kashmir but probably not northwestern Kashmir. It is open to further study whether this movement was related to the Late Palaeozoic/Early Mesozoic rifting of Gondwana's northern margin or to Greater India's Early Tertiary collision with southern Asia. The highly speculative nature of the above interpretation urges us to illucidate a possible alternative hypothesis for the origin of the fourth component, which accords with observations made elsewhere in the Himalaya. The generally east-west aligned shallow inclination directions tend to follow the general tectonic grain of the region. Comparable structurally aligned secondary components have been observed also in Ladakh [36] and in Nepal [3], in which cases a low-temperature oxidation phase with a direction preferentially aligned towards the foliation plane has been suggested [3,36]. This hypothesis may be applicable to the present case as well.

The Upper and Middle Triassic carbonates sampled at Khrew and the Lower Triassic and

Upper Permian carbonates sampled at Barsu show predominantly normal polarities. The lowermost sampled member A of the Zewan Beds (Barsu) thus postdates the Permo-Carboniferous (Kiaman) reversed polarity interval [37]. The stratigraphy of the Barsu section was studied in great detail [14,38] and a Punjabian age is well established for the A-member of the Zewan beds. This suggests that the upper age limit for the Kiaman interval may predate the Lower to Upper Tatarian time boundary which is generally agreed upon on basis of data from the U.S.S.R. [37,39,40].

6. Conclusions

Comparison of secondary (Middle to Late Tertiary) and primary (Late Permian and Triassic) magnetization components observed in the Palaeozoic to Lower Mesozoic platform sequence of Kashmir with the Indo-Pakistan APWP indicates that the region has undergone large-scale counterclockwise and clockwise rotations relative to the Indian shield region.

(A) Secondary magnetic components of Middle to Late Tertiary age show reversed-normal-reversed groupings of directions which for central and eastern Kashmir are distinctly offset with respect to the peninsular region. These results suggest a clockwise rotational movement of the Panjal Nappe with respect to the peninsula over approximately 45° since Middle Tertiary times.

(B) Primary magnetic directions of Late Permian and Triassic age show a 20–30° cumulative counterclockwise offset with respect to results from peninsular Indo-Pakistan. This indicates a 65–75° counterclockwise rotation of post-Middle to -Late Triassic age. This tectonic complexity reduces the potential of the here obtained primary magnetic directions (Triassic, Late Permian and probably Early Carboniferous, Siluro-Ordovician and Middle to Early Ordovician) for extension of the Indo-Pakistan APWP backwards in the Palaeozoic.

A consistent predominance of normal polarity directions within the Upper Permian/Triassic carbonate succession, suggests a pre-Punjabian upper time boundary for the Permo-Carboniferous (Kiaman) reversed polarity interval. This is earlier than the Late to Early Tatarian time boundary observed in the U.S.S.R.

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